MIL-H-44257
1 December 1986

MILITARY SPECIFICATION

HOMINY GRITS, WITH BACON AND CHEESE SAUCE, SMOKE FLAVORING ADDED, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers hominy grits with bacon and cheese sauce, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage: General Specification for

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipping and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standards for Grades of Butter
U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)

(Application for copies should be addressed to the Dairy Standardization Section, Dairy Division, Room 2750-S, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, P.O. Box 2109, Novato, CA 94947.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Hominy grits. Hominy grits shall be white, degermed, debrowned, corn. The hominy grits shall have a relatively bland flavor and a mild cereal aroma. The hominy grits shall comply with the standards of identity.

3.2.2 Bacon pieces, cooked. Cooked bacon pieces shall be a USDA inspected product produced from cured bacon with added smoke flavoring. The bacon pieces shall be free flowing and shall posses the color, odor, and flavor of properly cured and smoked bacon with no evidence of off-flavors and odors. The bacon pieces shall comply with the following analytical requirements: moisture content 36.7 $\pm$ 6.0 percent, fat content 25.1 $\pm$ 3.0 percent, and salt content 7.5 percent (maximum). The particle size shall be such that on a US Standard Series sieve, the following percentages (approximate) are retained:

US Standard Series Sieve	Percentage retained (approximate)
0.265	16.4
0.250 inch (US Standard 1/4)	29.0
0.187 inch (US Standard No. 4)	30.6
0.0937 inch (US Standard No. 8)	23.0

The bacon pieces shall be vacuum packaged in oxygen impermeable bags which have been backflushed with nitrogen. The packaging shall provide protection to the product from the effects of light. The packaged product shall be stored and shipped under refrigerated conditions. Time from manufacture to use of the bacon shall not exceed 120 days.

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3.2.3 Butter. Butter shall be lightly salted and shall be U.S. Grade A or better conforming to the U.S. Standards for Grades of Butter.

3.2.4 Cheese, pasteurized process, dehydrated. The dehydrated pasteurized process cheese shall be derived from fresh pasteurized process American cheese complying with the Standards of Identity for Pasteurized Process Cheese. The dehydrated cheese shall be produced by spray drying. The spray-dried product shall contain not less than 46 percent fat, not less than 32 percent protein, not more than 4 percent moisture, and not more than 1 percent carbohydrate. The dehydrated cheese shall be of a uniform light to medium orange color, shall be uniform in appearance and composition, and shall possess a characteristic pasteurized process cheese flavor and aroma. The product shall be free flowing, free from lumps that do no fall apart under light finger pressure, and free from specks of burnt particles.

3.2.5 Cheese, cheddar, sharp, dehydrated, white. Dehydrated cheddar cheese shall be derived from sharp, aged cheddar cheese. The dehydrated cheese shall be white in color, uniform in composition and appearance, free from burnt specks, and free from lumps that do not fall apart under light finger pressure. The cheese shall posses a characteristic flavor and odor. The dehydrated cheese shall contain not more than 4.0 percent moisture, not more than 5.0 percent salt, and not less than 45.0 percent milkfat.

3.2.6 Cheese, cheddar, sharp , dehydrated, yellow. Dehydrated cheddar cheese shall be derived from sharp, aged cheddar cheese. The dehydrated cheese shall be uniform in composition and appearance, free from burnt specks, and free from lumps that do not fall apart under light finger pressure. The cheese shall possess a characteristic flavor and odor and the color shall be within the range of No. 5 to No. 8 level when compared to the National Cheese institute's Cheese Color Reference Standards. The dehydrated cheese shall contain not more than 4.0 percent moisture, not more than 5.0 percent salt, and not less than 45.0 percent milkfat.

3.2.7 Milk, nonfat, dry, (low-heat). Nonfat dry milk shall be U.S. Extra Grade, Low-Heat as defined in the U.S. Standards for Grades of Nonfat Dry Milk (Spray Process).

3.2.8 Onion powder. Onion powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.9 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner coverings have been removed. The ground pepper shall have a characteristic penetrating odor, a hot, biting, pungent flavor, and a light color. The ground white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

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3.2.10 Salt. Salt shall be non-iodized, white, refined sodium chloride with or without anti-caking agents,

3.2.11 Sugar, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet, or a combination of both and shall comply with MIL-STD-900.

3.2.12 Water. Water used for formulation, ice making, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.13 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of the product. The hominy grits with bacon and cheese sauce shall be formulated and prepared as follows:

Ingredient	Percent by weight
Water	65.28
Bacon pieces	12.00
Hominy grits	11.50
Cheese, pasteurized process, dehydrated, American	5.65
Butter	3.30
Cheese, cheddar, sharp, dehydrated, yellow	0.75
Cheese, cheddar, sharp dehydrated, white	0.75
Salt 1/	0.15
Milk, nonfat, dry	0.35
Sugar	0.18
Onion powder	0.05
Pepper, white	0.04

1/ The total amount of salt in the formulation shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

a. The dry ingredients and the water shall be combined in a steam jacketed kettle and heated to boiling while thoroughly mixing. The mixture shall be allowed to cook for approximately 5 minutes or until the mixture is just thick enough to suspend the bacon pieces.

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b. The butter shall be added and thoroughly mixed until the butter has completely melted and dispersed throughout the mixture.

c. The bacon pieces shall be uniformly combined with the mixture to form the product.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The product shall be maintained in the temperature range of 150° to 180°F and filled into the tray pack cans within 4 hours after preparation.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. The temperature of the product at the time of filling shall be in the temperature range of 150°F to 180°F.

b. Immediately after filling, each can shall be hermetically sealed under a vacuum of 5 to 8 inches of mercury as determined by presence of visible concavity of the can lid.

c. The filled and sealed tray pack cans shall be in the retort process within 1 hour after being sealed.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at a temperature of 240°F or below until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be not foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, moldy, or sulfurous.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 102 ounces.

e. No individual can shall contain less than 100 ounces of product.

f. The salt content of any individual tray pack shall be not less than 0.5 percent nor greater than 1.3 percent.

g. The average fat content shall not be greater than 10.0 percent.

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h. No individual tray pack shall have a fat content greater than 12.0 percent.

i. Grits particles shall be tender but shall possess a tender to slightly grainy texture.

j. Bacon pieces shall be uniformly distributed through the product.

k. Free liquid in a can shall weigh not more than 1 ounce.

l. The product shall show no evidence of excessive heating (materially darkened or scorched.).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The bacon component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5 Quality conformance inspection.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, US Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 102 ounces or if the net weight of any individual can is less than 100 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated 35 minutes in boiling water, opened, and inspected for the defects listed in table II.

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TABLE I. Double sampling plan for product inspection. 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3200	8	---	0	2
	8	16	1	2
3201 to 35000	13	---	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	Grits particles do not posses a tender to slightly grainy texture.
102	Bacon pieces not uniformly distributed throughout the product.
103	Product shows evidence of excessive heating (materially darkened or scorched).
104	Free liquid in an individual can weighs more than 1 ounce. 3/

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.

3/ To determine the weight of free liquid, the can shall be opened and with the lid held in place the free liquid shall be decanted into a tared container allowing 1 minute for draining to occur. Report results of free liquid weight to nearest 0.1 ounce.

4.5.3.3 Fat and salt content testing. Nine tray packs shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis
- Three for submission to the contractor
- Three for retention by an AMS agent as a reserve sample.

The three tray packs for laboratory analyses shall be individually tested for fat and salt content in accordance with the Official Method of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, section: Meat, except that preparation of the samples shall be as follows: The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. Report test results to nearest 0.1 percent. The lot shall be rejected if:

a. The average fat content of the three tray packs is greater than 10.0 percent.
b. The fat content of any individual tray pack is greater than 12.0 percent.
c. The salt content of any individual tray pack is less than 0.5 percent or greater than 1.3 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of non-compliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers except that inspection for labeling shall be in accordance with MIL-L-1479 (see 5.4).

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with 21 CFR, Part 113, Subpart D and the can manufacturer's requirement.

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4.5.6 Vacuum examination. The sampling plan shall be as specified in table III. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge from bead to bead and verify that the distance from the straight edge to the lid is greater at the middle of the can than near the bead. The inspection lot shall include only tray packs produced in a single day on a single sealing machine.

TABLE III. Sampling plan for can vacuum examination

Lot size (cans)	Sample size (cans)	Acceptance number	Rejection number
0 to 1,200	32	0	1
1,201 to 3,200	50	0	1
3,201 to 10,000	80	1	2

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 102 ounces of the food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated.

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Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.1.1 When specified (see 6.1), the base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 98 pounds per base box. The body design shall be reinforced on the bottom with six longitudinal beads or ribs 0.050 ± 0.010 inches deep. The beads shall be 9 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the bottom. The end design shall be reinforced with eight longitudinal beads 0.018 ± 0.004 inches deep. The beads shall be 11 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the end. The center four beads on the end design shall be interrupted to provide a 5-inch by 5-inch flat center panel for labeling. The apex of all these beads shall be outward. The interior and exterior can coating requirements shall be as specified in 5.1.1.

5.1.2 Level C. The product shall be preserved as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a fiberboard box, constructed and closed in accordance with style HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The inside box dimensions shall be 12-3/4 inches in length by 10-1/2 inches in width by 8-1/2 inches in depth. The cans shall be packed flat, four in depth within the box, with the first two cans placed with ends down and the next two cans with ends up. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The dimensions of the fiberboard pads shall be 11-1/4 inches by 9-1/4 inches (± 1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F load.

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5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1 except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The inside box dimensions shall be 12-3/4 inches in length by 10-1/2 inches in width by 8-1/2 inches in depth. The cans shall be packed flat, four in depth within the box, with the first two cans placed with ends down and the next two cans with ends up. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade 275 fiberboard. The dimensions of the fiberboard pads shall be 11-1/4 inches by 9-1/4 inches (±1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping except for type II, class F loads..

5.4 Labeling. Each package shall be labeled in accordance with MIL-L-1497. In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.5.4 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge sealed can in boiling water. Simmer gently for 35 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. CAUTION: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.
WARNING: Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 18 servings of 2/3 cup each.

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5.5 Marking

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, when tray pack cans are fabricated from 90 pounds per base box, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. When tray pack cans are to be fabricated from 98 pounds per base box (see 5.1.1.1).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Canned foods
Hominy grits
Tray pack

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0568

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-44257	Hominy Grits, With Bacon and Cheese Sauce Smoke Flavoring Added, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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